

Definite Descriptions in First-Order Modal and Temporal Logics

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Definite Descriptions in KR Languages - DefDesKR
Half-Day Tutorial @ KR 2026

First-Order Modal Logics with Non-Rigid Designators and Counting

First-Order Modal and Temporal Logics

Extensions of First-Order Logic (FOL) with modal/temporal operators

- Full logics (highly) **undecidable**: contain FOL
- Quest for **decidable fragments**: generalisation of modal/temporal DLs

Non-Rigid Designators and Counting (NRDC) Features

- **equality/counting**;
- **non-rigid** and possibly **non-denoting constants**
- **definite descriptions**

First-Order Modal Logic with DDs: Background and Challenges

The Bad

Modal extensions of **decidable FO fragments** are typically **undecidable**, e.g.:

- Monadic fragment of FO **decidable**
- Monadic fragment of FOMLs K_n and $S5_n$, with $n \geq 1$, **undecidable**

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Monodic fragments: modalities applied only to formulas with ≤ 1 **free variable**

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- . . . but rely on the absence of NRDC features!

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The Ugly

Mostly **negative results** on computational behaviour of fragments with NRDC features

- from product modal logics & fragments of FO modal/temporal logics with counting

First-Order Modal Logic with DDs: Recent Results

Investigation of **decidability** and **complexity boundaries**
for **monodic fragments** with **NRDC** features

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- **Decision problems**: validity and global consequence

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- **Domains**: **constant** or **expanding**
- **Decision problems**: **validity** and **global consequence**
 - Global consequence **not reducible** to validity in general, but...
 - ...**reducible** on frames w/: single equivalence relation; transitive closure; or time flows

$\mathcal{Q}^=ML_\iota$, the First-Order Modal Language with NRDC

Terms and Formulas

$\mathcal{Q}^=ML_\iota$ **terms** and **formulas** defined by mutual induction:

$$\tau ::= x \mid c \mid \iota x. \varphi$$

$$\varphi ::= P(\tau_1, \dots, \tau_m) \mid \tau_1 = \tau_2 \mid \neg \varphi \mid (\varphi_1 \wedge \varphi_2) \mid \exists x \varphi \mid \Diamond_a \varphi$$

- **variables** $x \in \text{Var}$, **constants** $c \in \text{Con}$, and **predicates** $P \in \text{Pred}$ (m -ary)
- finite set of **modalities** $a \in A$
- **definite descriptions** $\iota x. \varphi$, read as “the x such that φ ”

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Notation

- $\varphi_1 \vee \varphi_2 := \neg(\neg\varphi_1 \wedge \neg\varphi_2)$, $\varphi_1 \rightarrow \varphi_2 := \neg\varphi_1 \vee \varphi_2$, $\varphi_1 \leftrightarrow \varphi_2 := (\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1)$
- $\forall x \varphi := \neg \exists x \neg \varphi$
- $\Box_a \varphi := \neg \Diamond_a \neg \varphi$
- Γ finite set of sentences (no free variables)

Semantics with Partial Interpretations

Partial Interpretation with Expanding Domains

$\mathfrak{M} = (\mathfrak{F}, \Delta, \cdot)$ where:

- $\mathfrak{F} = (W, \{R_a\}_{a \in A})$ **frame** with worlds W ($\neq \emptyset$) and accessibility relations R_a
- Δ function assigning **domain** Δ_w ($\neq \emptyset$) to each $w \in W$ s.t. $\Delta_w \subseteq \Delta_v$ when wR_av
- $\cdot$ function mapping each $w \in W$ to **partial FO interpretation** $\mathfrak{M}(w)$ with:
 - $P^{\mathfrak{M}(w)} \subseteq \Delta_w^m$, for each m -ary predicate $P \in \text{Pred}$ (total on predicates)
 - $c^{\mathfrak{M}(w)} \in \Delta_w$, for **some constant** symbols $c \in \text{Con}$ (**partial** on constants)

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Designation, Total Interpretations, Constant Domains, Assignments

- c **designates at** w : $c^{\mathfrak{M}(w)}$ is defined
- **total interpretation**: all constants designate at all worlds
- **constant domains**: $\Delta_w = \Delta_v$ for all $w, v \in W$
- **assignment at** w : function α from Var to Δ_w
- **x -variant** of α at w : assignment α' at w that can differ from α only on x

Term Interpretation and Satisfaction

Value of Terms

$$\tau^{\mathfrak{M}(w), \alpha} = \begin{cases} \alpha(x), & \text{if } \tau = x \in \text{Var} \\ c^{\mathfrak{M}(w)}, & \text{if } \tau = c \in \text{Con} \text{ and } c^{\mathfrak{M}(w)} \text{ **defined**} \\ \alpha'(x), & \text{if } \tau = \iota x. \varphi \text{ and } \mathfrak{M}, w \models^{\alpha'} \varphi \text{ for **exactly one** } x\text{-variant } \alpha' \text{ of } \alpha \\ \text{undefined}, & \text{otherwise} \end{cases}$$

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Satisfaction Relation

- $\mathfrak{M}, w \models^{\alpha} P(\tau_1, \dots, \tau_m)$ iff all $\tau_i^{\mathfrak{M}(w), \alpha}$ **defined** and $(\tau_1^{\mathfrak{M}(w), \alpha}, \dots, \tau_m^{\mathfrak{M}(w), \alpha}) \in P^{\mathfrak{M}(w)}$
- $\mathfrak{M}, w \models^{\alpha} \tau_1 = \tau_2$ iff both $\tau_i^{\mathfrak{M}(w), \alpha}$ **defined** and $\tau_1^{\mathfrak{M}(w), \alpha} = \tau_2^{\mathfrak{M}(w), \alpha}$
- $\mathfrak{M}, w \models^{\alpha} \exists x \varphi$ iff there exists x -variant α' with $\mathfrak{M}, w \models^{\alpha'} \varphi$
- $\mathfrak{M}, w \models^{\alpha} \Diamond_a \varphi$ iff there exists $v \in W$ such that $wR_a v$ and $\mathfrak{M}, v \models^{\alpha} \varphi$

Truth, Validity, and Global Consequence

Truth, Satisfaction

- φ **true in** $\mathfrak{M}$, $\mathfrak{M} \models \varphi$: φ satisfied under every assignment at every world of $\mathfrak{M}$
- φ **satisfied in** $\mathfrak{M}$: φ satisfied under some assignment at some world of $\mathfrak{M}$
- Γ **true in** $\mathfrak{M}$, $\mathfrak{M} \models \Gamma$: every sentence in Γ is true in $\mathfrak{M}$

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Validity, Satisfiability, Global Consequence

$\mathcal{C}$ class of frames (see below). Formula φ

- $\mathcal{C}$ -**valid**: φ true in every interpretation $\mathfrak{M}$ based on a frame $\mathfrak{F} \in \mathcal{C}$
- $\mathcal{C}$ -**satisfiable**: φ satisfied in some interpretation $\mathfrak{M}$ based on a frame $\mathfrak{F} \in \mathcal{C}$
- **global $\mathcal{C}$ -consequence of theory Γ** : φ true in any interpretation $\mathfrak{M}$ based on a frame in $\mathcal{C}$ such that $\mathfrak{M} \models \Gamma$

Classes of Frames for Modal and Temporal Logics

K_n and $S5_n$ Frames

- K_n : class of all frames with n **arbitrary** accessibility relations
- $S5_n$: class of frames with n **equivalence relations**; $S5 := S5_1$

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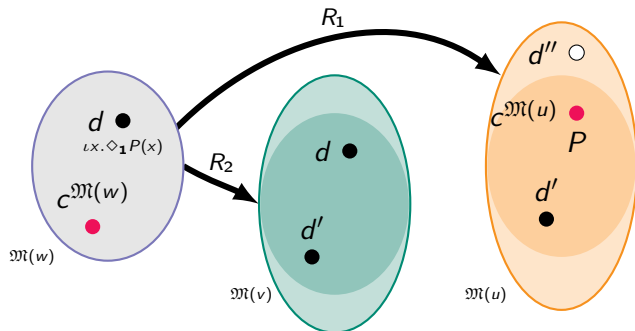
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Linear Temporal Frames

- $LTL^\diamond$: $\{(\mathbb{N}, <)\}$, with standard **strict linear order** $<$ (interpreting $\diamond$)
- LTL^f : $\{(\{0, \dots, n\}, <) \mid n \in \mathbb{N}\}$, with $<$ restricted to **finite** $\{0, \dots, n\}$
- LTL : $\{(\mathbb{N}, <, S)\}$, with **successor** relation $S = \{(i, i+1) \mid i \in \mathbb{N}\}$ (interpreting $\bigcirc$)
- LTL^f : $\{(\{0, \dots, n\}, <, S) \mid n \in \mathbb{N}\}$, with $<$ and S restricted to **finite** $\{0, \dots, n\}$

Examples

Partial Interpretation with Non-Rigid Designators



$$\exists x(x \neq c \wedge \Diamond_1(x = c))$$

unsatisfiable if constants are **rigid**

Examples

Vulcan and Venus

- “It is conceivable *that* Vulcan is the planet orbiting between Sun and Mercury”:

$\Diamond(\text{vulcan} = \iota z.\text{OrbitsBetween}(z, \text{sun}, \text{mercury}))$

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- “Even though such a planet does not exist”:

$$\neg \exists x(x = \text{vulcan}) \wedge \neg \exists x(x = \iota z.\text{OrbitsBetween}(z, \text{sun}, \text{mercury}))$$

i.e., neither ‘vulcan’ nor ‘ $\iota z.\text{OrbitsBetween}(z, \text{sun}, \text{mercury})$ ’ designate in this world

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- “It is known *of* the planet orbiting between Mercury and Earth that it is Venus”:

$$\exists x(x = \iota z.\text{OrbitsBetween}(z, \text{mercury}, \text{earth}) \wedge \Box(x = \text{venus}))$$

in **S5** frames, this implies that ‘venus’ is rigid

Monodic Fragments

Monodic Fragment

Set $Q_{\square}^{\equiv}ML_{\iota}$ of **monodic** formulas:

- every subformula of the form $\Diamond_a\psi$ has **at most one free variable**.

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Minimal Sub-Fragment

- $Q_{\square}^1 ML_{\iota}$: **One-variable** fragment with ≤ 1 -ary predicates (and equality)

Maximal Sub-Fragments

- $C_{\square}^2 ML_{\iota}$: **Two-variable** fragment with **counting** quantifiers and ≤ 2 -ary predicates
- $GF_{\square}^{\equiv} ML_{\iota}$: **Guarded** fragment with equality

One-Variable Fragment $Q^1=ML_\ell$

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$Q^1=ML_\ell$ **terms** and **formulas** built with:

- **one variable** only
- predicates of **arity at most one** (plus equality)
- constants and definite descriptions

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Remarks

- Underpinned by FO^1 with equality and constants
- All formulas trivially monodic
- Variants extensively studied as product modal logics

Two-Variable Fragment with Counting $C_{\square}^2 ML_{\iota}$

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$C_{\square}^2 ML_{\iota}$ terms and formulas built with:

- **two variables**
- **counting quantifiers** $\exists^{\geq k} x$, $k \geq 0$
- predicates of **arity at most two** (including equality)
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Counting Quantifier

$\mathfrak{M}, w \models^a \exists^{\geq k} x \varphi$ iff $\mathfrak{M}, w \models^{a'} \varphi$ for at least k distinct x -variants a'

Guarded Fragment $\text{GF}_{\square}^{\equiv} \text{ML}_{\iota}$

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- constants and **closed** definite descriptions $\iota x. \chi(x)$, $\chi(x)$ with ≤ 1 free variable x

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Remark

Closed definite descriptions necessary for decidability even without modalities

- $\forall x F(x, \iota y. F(x, y))$ ensures F is a **function**
- guarded fragment with **functionality** is **undecidable**

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$\exists x \varphi(x)$ can still be expressed

- equivalent to $\exists x ((x = x) \wedge \varphi(x))$, with $x = x$ as its guard.

Decision Problems

Validity and Global Consequence Decision Problems

For fragment $\mathcal{L}$ and frame class $\mathcal{C}$

- **$\mathcal{C}$ -validity in $\mathcal{L}$:** Is φ valid on all interpretations based on $\mathcal{C}$ -frames?
- **global $\mathcal{C}$ -consequence in $\mathcal{L}$:** Is φ true in all interpretations based on $\mathcal{C}$ -frames that make theory Γ true?

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Problem Qualifiers

- **total $\mathcal{C}$ -validity:** restriction to total interpretations
- with **constant** domains, with **expanding** domains

Overview of Results

frames $\mathcal{C}$	dom.	$\mathcal{C}$ -validity			global $\mathcal{C}$ -consequence		
		$Q^1=ML_\ell$	$C^2_{\Box}ML_\ell$	$GF^=_{\Box}ML_\ell$	$Q^1=ML_\ell$	$C^2_{\Box}ML_\ell$	$GF^=_{\Box}ML_\ell$
S5	=	coNEXP	coNEXP	2EXP	coNEXP	coNEXP	2EXP
S5 $_n$, $n \geq 2$	=	coNEXP	coNEXP	2EXP	undecidable		
K $_n$	=	coNEXP	coNEXP	2EXP	undecidable		
	$\subseteq$	PSPACE	coNEXP	2EXP	?		
K $_{*n}$, LTL $^{(\diamond)}$	=	Σ^1_1					
	$\subseteq$	undecidable					
Kf $_{*n}$, LTLf $^{(\diamond)}$	=	undecidable					
	$\subseteq$	decidable, Ackermann-hard					

Preliminary Observations

Elsewhere Operator

- Reductions between our one-variable fragment with **non-rigid constants** and $Q^{1\neq}ML$, the one-variable fragment with **difference/elsewhere** quantifier $\exists^{\neq}x$:

$$\varphi ::= P(x) \mid \neg\varphi \mid (\varphi \wedge \varphi) \mid \exists x \varphi \mid \exists^{\neq}x \varphi \mid \Diamond_a \varphi$$

$$\mathfrak{M}, w \models^a \exists^{\neq}x \varphi \quad \text{iff} \quad \mathfrak{M}, w \models^{a'} \varphi, \text{ for some } x\text{-variant } a' \neq a \text{ at } w$$

[Hampson & Kurucz, AiML 2012; Hampson & Kurucz, ACM TOCL 2015]

- Lift results from propositional case: “**difference** $\equiv$ **nominals** + **universal modality**”

[Gargov & Goranko, JPL, 1993]

- from $Q^{1=}ML_c$ to $Q^{1\neq}ML$: $x = c \rightsquigarrow Q_c(x) \wedge \neg \exists^{\neq}x Q_c(x)$
- from $Q^{1\neq}ML$ to $Q^{1=}ML_c$: $\exists^{\neq}x \psi \rightsquigarrow \exists x \psi(x) \wedge (x = c_\psi \rightarrow \exists x (\neg(x = c_\psi) \wedge \psi(x)))$

Preliminary Observations

Getting Rid of Definite Descriptions...

Normalise and **eliminate definite descriptions**: replace definite descriptions $\iota x.\psi(x, \mathbf{y})$ with “atomic surrogates” $\iota x.P_\psi(x, \mathbf{y})$; then, replace atoms $\alpha(\iota x.Q(x, \mathbf{y}), \tau)$ with their “Russell’s paraphrase”:

- in $Q_{\square}^{\equiv}ML_{\iota}$, $\exists x(\alpha(x, \tau) \wedge Q(x, \mathbf{y}) \wedge \forall x'(Q(x', \mathbf{y}) \rightarrow x' = x))$
- in $C_{\square}^2ML_{\iota}$ $\exists x(\alpha(x, \tau) \wedge Q(x, \mathbf{y})) \wedge \exists^{\equiv 1}xQ(x, \mathbf{y})$
- in $Q^1=ML_{\iota}$ and $GF_{\square}^{\equiv}ML_{\iota}$ $\alpha(c_{\iota x.Q(x)}, \tau) \wedge Q(c_{\iota x.Q(x)}) \wedge \forall x (Q(x) \rightarrow x = c_{\iota x.Q(x)})$

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- in $C^2_\square ML_\iota$ $\exists x(\alpha(x, \boldsymbol{\tau}) \wedge Q(x, \mathbf{y})) \wedge \exists^{=1} x Q(x, \mathbf{y})$
- in $Q^1=_\square ML_\iota$ and $GF^=_\square ML_\iota$ $\alpha(c_{\iota x.Q(x)}, \boldsymbol{\tau}) \wedge Q(c_{\iota x.Q(x)}) \wedge \forall x (Q(x) \rightarrow x = c_{\iota x.Q(x)})$

...And Other Useful Reductions

- **Reduce partial to total** interpretations, and **viceversa**:
 - Partial $\rightsquigarrow$ Total: replace constants with propositional letter designation switch

$$P(\tau_1, \dots, \tau_m) \rightsquigarrow \bigwedge_{c_i \in \{\tau_1, \dots, \tau_m\}} p_{c_i} \wedge P(\tau_1, \dots, \tau_m)$$

- Total $\rightsquigarrow$ Partial: add “**designation axioms**” for $c \rightsquigarrow \exists x(x = c)$
- **Reduce expanding to constant** domains

Overview of Decidability Results

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Main Ideas for Decidability

From Models to Quasimodels

Adapt **quasimodel** machinery for NRDC features

- Use **multisets** of types and runs to handle counting
- ... but they can still require **infinite branching**

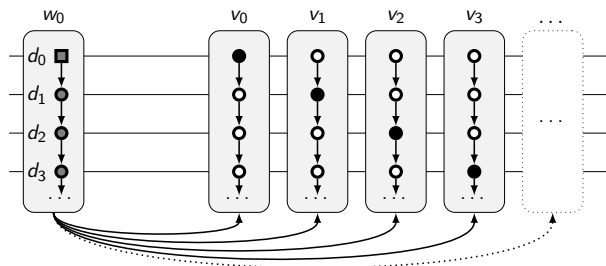


Figure: $\forall x \exists^{\leq 1} y P(x, y) \wedge \forall x \exists^{\leq 1} z P(z, x) \wedge \exists x \neg \exists z P(z, x) \wedge \forall x \Diamond_a A(x) \wedge \Box_a \exists^{\leq 1} x A(x)$

Main Ideas for Decidability

From Quasimodels to Weak Quasimodels

Introduce **weak quasimodels** with **relaxed saturation conditions**

- Saturate only a representative for each multiset of types
- Avoid infinite branching of quasimodels due to counting

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- Avoid infinite branching of quasimodels due to counting

Bounded-Size Weak Quasimodels

Use **bounded-size** weak quasimodels to decide satisfiability

- 1-variable and guarded fragments: use weak quasimodels directly
- 2-variable with counting: introduce weak pre-quasimodels + (in)equalities encoding constraints in Presburger Arithmetic with infinity

Main Ideas for Decidability

From Quasimodels to Weak Quasimodels

Introduce **weak quasimodels** with **relaxed saturation conditions**

- Saturate only a representative for each multiset of types
- Avoid infinite branching of quasimodels due to counting

Bounded-Size Weak Quasimodels

Use **bounded-size** weak quasimodels to decide satisfiability

- 1-variable and guarded fragments: use weak quasimodels directly
- 2-variable with counting: introduce weak pre-quasimodels + (in)equalities encoding constraints in Presburger Arithmetic with infinity

Simpler Expanding Domain Cases

- validity/global consequence in fragments $\left\{ \begin{array}{l} \text{with transitive closure \& no infinite chain} \\ \text{on finite temporal frames} \end{array} \right.$
- K_n -validity in one-variable fragment

Overview of Undecidability Results

frames $\mathcal{C}$	dom.	$\mathcal{C}$ -validity			global $\mathcal{C}$ -consequence		
		$Q^1=ML_\iota$	$C^2_{\Box}ML_\iota$	$GF^=_{\Box}ML_\iota$	$Q^1=ML_\iota$	$C^2_{\Box}ML_\iota$	$GF^=_{\Box}ML_\iota$
S5	=	coNEXP	coNEXP	2EXP	coNEXP	coNEXP	2EXP
S5_n, $n \geq 2$	=	coNEXP	coNEXP	2EXP	undecidable		
K_n	=	coNEXP	coNEXP	2EXP	undecidable		
	$\subseteq$	PSPACE	coNEXP	2EXP	?		
K_{*n}, LTL($\diamond$)	=	Σ^1_1					
	$\subseteq$	undecidable					
Kf_{*n}, LTLf($\diamond$)	=	undecidable					
	$\subseteq$	decidable, Ackermann-hard					

Main Ideas for Undecidability

Main Ideas for Undecidability

- For global consequence on K_n ($n \geq 1$) and $S5_n$ ($n \geq 2$) frames, reduce:
 - **undecidable products** to one-variable fragment with **elsewhere quantifier** $\exists^{\neq} x$
 - the latter to our one-variable fragment with **non-rigid constants**
- For validity/global consequence with **transitive closure** & on **temporal** (infinite with constant/expanding domains, or finite with constant domains) frames, reduce
 - **undecidable one-variable FOTL with counting** to our one-variable fragments on temporal (infinite or finite, resp.) frames
 - the latter to one-variable fragments with **transitive closure** (with or without infinite chains, resp.)

Other Approaches to First-Order Modal Logic with Non-Rigid Terms

An Ocean of Possibilities

- Syntax: scope disambiguation for non-rigid terms and modal operators with predicate λ -abstraction / typed languages; DDs as referring expressions for time points: “the first time φ was true”
- Proof theory: axiom systems, tableaux, sequent calculi
- Semantics: double-domain, counterparts, modal categories, sheaves, . . .

[Garson, *HPL*, 2001; Corsi, *JSL* 2002; Shehtman, *AiML* 2006, Braüner & Ghilardi, *HML*, 2007; Awodey & Kishida, *AiML* 2012; Indrzejczak, *AiML* 2020; Fitting & Mendelsohn, *FOML*, 2023; Indrzejczak & Zawidzki, *Synthese* 2023; Orlandelli, *JPL* 2024, Shkatov & Shehtman, *ESSLLI* 2025; Ghilardi & Marquès, *JSL* 2025]